

DAILY DRILLING REPORT SET

SYN-Assam-Duliajan-009 (W009) — SYN-Assam-Duliajan

1. REPORT SET IDENTIFICATION

Well ID	W009
Well Name	SYN-Assam-Duliajan-009
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude / Longitude	27.396274 N / 95.396813 E
Rig	SYN-RIG-N1600-02
Spud Date	17-Dec-2009
Final Report Date	27-Dec-2009
Daily Reports in Set	11 (Day 01 to Day 11)
Total Depth Reached	3,180 m
Primary Formation	Sandstone
Events Reported in Set	6

2. DEPTH PROGRESSION SUMMARY

Day	Date	From (m)	To (m)	Drilled (m)	Formation	Events
01	17-Dec-2009	0	500	500	Claystone	Cementing Issue
02	18-Dec-2009	500	840	340	Claystone / Shale	Cementing Issue
03	19-Dec-2009	840	1,230	390	Shale	—
04	20-Dec-2009	1,230	1,585	355	Shale	—
05	21-Dec-2009	1,585	1,905	320	Shale / Sandstone	Wellbore Instability
06	22-Dec-2009	1,905	2,100	195	Sandstone	Mud Loss, Mud Loss
07	23-Dec-2009	2,100	2,310	210	Sandstone	—
08	24-Dec-2009	2,310	2,565	255	Sandstone	—
09	25-Dec-2009	2,565	2,750	185	Sandstone	Mud Loss
10	26-Dec-2009	2,750	2,970	220	Sandstone / Limestone	—
11	27-Dec-2009	2,970	3,180	210	Limestone	—

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). Depths, dates, parameters and events are generated from the prototype datasets and are not operational records of any company.

DAILY DRILLING REPORT — DAY 01	W009 · 17-Dec-2009
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Well ID: W009
Well Name: SYN-Assam-Duliajan-009
Report Date: 2009-12-17
Day Number: 1
Start Depth: 0 m
End Depth: 500 m
Drilled Interval: 500 m
Formation: Claystone
Mud Weight: 9.43 ppg
ROP: 33.7 m/hr
RPM: 137
WOB: 7.6 klbf
Events: Cementing Issue at 500 m (Medium)
NPT: 7.4 hours

Operations performed.

- Rig up completed and well spudded at 17-Dec-2009. Drilled 17-1/2" hole from surface to 500 m.
- Circulated and conditioned mud; hi-vis sweep circulated bottoms-up.
- Cementing Issue handled at 500 m — 7.4 hr non-productive time. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage.

Geological observations.

Cuttings throughout the interval are claystone, consistent with the 0–743 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 9.43 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Casing/cementing equipment inspected after the event; cement unit and pressure recorders checked and found serviceable.

Events reported this day.

Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0053	500	Claystone	Cementing Issue	Medium	7.4	Partial non-productive time; drilling suspended while the well was treated and conditioned.

End-of-day status.

Depth 500 m in Claystone. Drilling ahead. Non-productive time this tour 7.4 hr.

DAILY DRILLING REPORT — DAY 02	W009 · 18-Dec-2009
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Well ID: W009
Well Name: SYN-Assam-Duliajan-009
Report Date: 2009-12-18
Day Number: 2
Start Depth: 500 m
End Depth: 840 m
Drilled Interval: 340 m
Formation: Claystone / Shale
Mud Weight: 9.96 ppg
ROP: 27.8 m/hr
RPM: 136
WOB: 10.8 klbf
Events: Cementing Issue at 592 m (Low)
NPT: 3.7 hours

Operations performed.

- Drilled 340 m of section from 500 m to 840 m (12.8 hr on bottom).
- Deviation survey taken; hole angle within tolerance.
- Pulled out of hole, ran 13-3/8" casing to 745 m and cemented as programmed (Lead 12.6 ppg / Tail 15.8 ppg, cement to surface). Waited on cement and pressure tested.
- Cementing Issue handled at 592 m — 3.7 hr non-productive time. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage.

Geological observations.

Formation change crossed during the tour: Claystone to 743 m, Shale to 1,783 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 9.96 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Casing/cementing equipment inspected after the event; cement unit and pressure recorders checked and found serviceable.

Events reported this day.

Prototype Training Record · Synthetic demonstration data

Page 2 of 7

DAILY DRILLING REPORT — DAY 03

W009 · 19-Dec-2009

Well ID: W009
Well Name: SYN-Assam-Duliajan-009
Report Date: 2009-12-19
Day Number: 3
Start Depth: 840 m
End Depth: 1230 m
Drilled Interval: 390 m
Formation: Shale
Mud Weight: 10.59 ppg
ROP: 17.0 m/hr
RPM: 131
WOB: 14.8 klb
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 390 m of Shale from 840 m to 1,230 m (22.5 hr on bottom).
- Slipped and cut drilling line; routine BOP function test carried out.

Geological observations.

Cuttings throughout the interval are shale, consistent with the 743–1,783 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 10.59 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

All rig equipment functioning normally; no downtime recorded.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 1,230 m in Shale. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 04

W009 · 20-Dec-2009

Well ID: W009
Well Name: SYN-Assam-Duliajan-009
Report Date: 2009-12-20
Day Number: 4
Start Depth: 1230 m
End Depth: 1585 m
Drilled Interval: 355 m
Formation: Shale
Mud Weight: 11.07 ppg
ROP: 16.2 m/hr
RPM: 123
WOB: 17.5 klb
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 355 m of Shale from 1,230 m to 1,585 m (22.5 hr on bottom).
- Circulated and conditioned mud; hi-vis sweep circulated bottoms-up.

Geological observations.

Cuttings throughout the interval are shale, consistent with the 743–1,783 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 11.07 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Solids-control equipment cleaned; shaker screens changed. No downtime.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 1,585 m in Shale. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 05						W009 · 21-Dec-2009
Well ID: W009 Well Name: SYN-Assam-Duliajan-009 Report Date: 2009-12-21 Day Number: 5 Start Depth: 1585 m End Depth: 1905 m Drilled Interval: 320 m Formation: Shale / Sandstone Mud Weight: 11.41 ppg ROP: 16.3 m/hr RPM: 122 WOB: 19.1 klbf Events: Wellbore Instability at 1879 m (High) NPT: 19.0 hours						
Operations performed. - Drilled 320 m of section from 1,585 m to 1,905 m (2.0 hr on bottom). - Made connections and circulated bottoms-up at intervals; shakers monitored. - Pulled out of hole, ran 9-5/8" casing to 1,785 m and cemented as programmed (Lead 12.9 ppg / Tail 15.8 ppg, TOC 630 m). Waited on cement and pressure tested. - Wellbore Instability handled at 1,879 m — 19.0 hr non-productive time. Mud inhibition and rheology increased, sweeps circulated to clear cavings and controlled back-reaming carried out; ROP limited through the unstable interval.						
Geological observations. Formation change crossed during the tour: Shale to 1,783 m, Sandstone to 2,817 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 11.41 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. Mud pump No. 2 packing changed during routine circulation; no lost time.						
Events reported this day.						
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0056	1,879	Sandstone	Wellbore Instability	High	19.0	Significant non-productive time; remedial operations and additional materials required.
End-of-day status. Depth 1,905 m in Shale. 9-5/8" casing cemented at 1,785 m; waiting on cement.						

DAILY DRILLING REPORT — DAY 06						W009 · 22-Dec-2009
Well ID: W009 Well Name: SYN-Assam-Duliajan-009 Report Date: 2009-12-22 Day Number: 6 Start Depth: 1905 m End Depth: 2100 m Drilled Interval: 195 m Formation: Sandstone Mud Weight: 11.82 ppg ROP: 15.7 m/hr RPM: 117 WOB: 21.0 klbf Events: Mud Loss at 1921 m (High); Mud Loss at 2068 m (Medium) NPT: 20.8 hours						
Operations performed. - Drilled 195 m of Sandstone from 1,905 m to 2,100 m (2.0 hr on bottom). - Circulated and conditioned mud; hi-vis sweep circulated bottoms-up. - Mud Loss handled at 1,921 m — 15.3 hr non-productive time. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet. - Mud Loss handled at 2,068 m — 5.5 hr non-productive time. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.						
Geological observations. Cuttings throughout the interval are sandstone, consistent with the 1,783–2,817 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Returns reduced during the tour; losses treated with LCM and mud weight trimmed. Loss volumes recorded on the daily loss sheet. Mud weight held at 11.82 ppg after treatment.						
Equipment / operational observations.						

DAILY DRILLING REPORT — DAY 07

W009 · 23-Dec-2009

Well ID: W009
Well Name: SYN-Assam-Duliajan-009
Report Date: 2009-12-23
Day Number: 7
Start Depth: 2100 m
End Depth: 2310 m
Drilled Interval: 210 m
Formation: Sandstone
Mud Weight: 12.00 ppg
ROP: 15.4 m/hr
RPM: 118
WOB: 22.3 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 210 m of Sandstone from 2,100 m to 2,310 m (22.5 hr on bottom).
- Slipped and cut drilling line; routine BOP function test carried out.

Geological observations.

Cuttings throughout the interval are sandstone, consistent with the 1,783–2,817 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 12.00 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Solids-control equipment cleaned; shaker screens changed. No downtime.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 2,310 m in Sandstone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 08

W009 · 24-Dec-2009

Well ID: W009
Well Name: SYN-Assam-Duliajan-009
Report Date: 2009-12-24
Day Number: 8
Start Depth: 2310 m
End Depth: 2565 m
Drilled Interval: 255 m
Formation: Sandstone
Mud Weight: 12.32 ppg
ROP: 15.8 m/hr
RPM: 117
WOB: 24.8 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 255 m of Sandstone from 2,310 m to 2,565 m (22.5 hr on bottom).
- Circulated and conditioned mud; hi-vis sweep circulated bottoms-up.

Geological observations.

Cuttings throughout the interval are sandstone, consistent with the 1,783–2,817 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 12.32 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

BOP drill carried out with the crew; response time satisfactory.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 2,565 m in Sandstone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 09						W009 · 25-Dec-2009														
Well ID: W009 Well Name: SYN-Assam-Duliajan-009 Report Date: 2009-12-25 Day Number: 9 Start Depth: 2565 m End Depth: 2750 m Drilled Interval: 185 m Formation: Sandstone Mud Weight: 12.59 ppg ROP: 15.6 m/hr RPM: 109 WOB: 24.9 klbf Events: Mud Loss at 2597 m (Medium) NPT: 6.1 hours																				
Operations performed. - Drilled 185 m of Sandstone from 2,565 m to 2,750 m (16.4 hr on bottom). - Made connections and circulated bottoms-up at intervals; shakers monitored. - Mud Loss handled at 2,597 m — 6.1 hr non-productive time. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.																				
Geological observations. Cuttings throughout the interval are sandstone, consistent with the 1,783–2,817 m unit logged in the offset wells of this cluster.																				
Mud / circulation observations. Returns reduced during the tour; losses treated with LCM and mud weight trimmed. Loss volumes recorded on the daily loss sheet. Mud weight held at 12.59 ppg after treatment.																				
Equipment / operational observations. All rig equipment functioning normally; no downtime recorded.																				
Events reported this day. <table><tr><th>Event ID</th><th>Depth (m)</th><th>Formation</th><th>Type</th><th>Severity</th><th>Hr</th><th>Operational Impact</th></tr><tr><td>E0055</td><td>2,597</td><td>Sandstone</td><td>Mud Loss</td><td>Medium</td><td>6.1</td><td>Partial non-productive time; drilling suspended while the well was treated and conditioned.</td></tr></table>							Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact	E0055	2,597	Sandstone	Mud Loss	Medium	6.1	Partial non-productive time; drilling suspended while the well was treated and conditioned.
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact														
E0055	2,597	Sandstone	Mud Loss	Medium	6.1	Partial non-productive time; drilling suspended while the well was treated and conditioned.														
End-of-day status. Depth 2,750 m in Sandstone. Drilling ahead. Non-productive time this tour 6.1 hr.																				

DAILY DRILLING REPORT — DAY 10						W009 · 26-Dec-2009
Well ID: W009 Well Name: SYN-Assam-Duliajan-009 Report Date: 2009-12-26 Day Number: 10 Start Depth: 2750 m End Depth: 2970 m Drilled Interval: 220 m Formation: Sandstone / Limestone Mud Weight: 13.15 ppg ROP: 10.4 m/hr RPM: 108 WOB: 29.2 klbf Events: None NPT: 0.0 hours						
Operations performed. - Drilled 220 m of section from 2,750 m to 2,970 m (22.5 hr on bottom). - Made connections and circulated bottoms-up at intervals; shakers monitored.						
Geological observations. Formation change crossed during the tour: Sandstone to 2,817 m, Limestone to 3,180 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 13.15 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. All rig equipment functioning normally; no downtime recorded.						
Events reported this day. None. Operations proceeded within the planned parameter envelope.						
End-of-day status. Depth 2,970 m in Limestone. Drilling ahead. No non-productive time this tour.						

DAILY DRILLING REPORT — DAY 11	W009 · 27-Dec-2009
Well ID: W009 Well Name: SYN-Assam-Duliajan-009 Report Date: 2009-12-27 Day Number: 11 Start Depth: 2970 m End Depth: 3180 m Drilled Interval: 210 m Formation: Limestone Mud Weight: 13.56 ppg ROP: 8.3 m/hr RPM: 97 WOB: 32.8 klbf Events: None NPT: 0.0 hours	
Operations performed. • Drilled 210 m of Limestone from 2,970 m to 3,180 m (16.5 hr on bottom). • Deviation survey taken; hole angle within tolerance. • Pulled out of hole, ran 7" casing to 3,180 m and cemented as programmed (Tail 15.8 ppg, TOC 1,640 m). Waited on cement and pressure tested. • Reached driller's total depth of 3,180 m. Circulated hole clean, logged and secured the well. Geological observations. Cuttings throughout the interval are limestone, consistent with the 2,817–3,180 m unit logged in the offset wells of this cluster. Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 13.56 ppg, properties held within programme; no losses or gains recorded. Equipment / operational observations. Solids-control equipment cleaned; shaker screens changed. No downtime. Events reported this day. None. Operations proceeded within the planned parameter envelope. End-of-day status. Depth 3,180 m in Limestone. 7" casing cemented at 3,180 m; waiting on cement.	